

Week 2: Mortality I

SOC6708 ADA

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Read in data

```
library(tidyverse)
library(here)
library(readxl)
library(janitor)
```

For this example we are using mortality in Canada by cause, 2012-2016. Available from [StatCan](#).

```
d <- read_csv(here("data", "CAN_age_cod.csv"))
head(d)
```

```
# A tibble: 6 x 5
  start_age cause          year number rate
  <dbl> <chr>          <dbl> <dbl> <dbl>
1      0 Total, all causes of death [A00-Y89] 2012  1818  483
2      0 Total, all causes of death [A00-Y89] 2013  1884  492
3      0 Total, all causes of death [A00-Y89] 2014  1794  471
4      0 Total, all causes of death [A00-Y89] 2015  1737  454
5      0 Total, all causes of death [A00-Y89] 2016  1741  450
6      0 Salmonella infections [A01-A02] 2012     0    0
```

Create a life table

Let's create a life table using the mortality rates from all causes of death in 2016. Get the data we need:

```
dl <- d %>%
  filter(year==2016, cause=="Total, all causes of death [A00-Y89]") %>%
  mutate(age = start_age, Mx = rate/100000) %>%
  select(age, Mx)
```

```
head(dl)
```

```
# A tibble: 6 x 2
```

	age	Mx
	<dbl>	<dbl>
1	0	0.0045
2	1	0.00019
3	5	0.00008
4	10	0.00013
5	15	0.00033
6	20	0.00053

We need to create columns:

- n
- ${}_n a_x$
- ${}_n q_x$
- ${}_n d_x$
- ${}_n L_x$
- T_x
- e_x

```
d_lt <- dl |>
  mutate(n = case_when(
    age==0 ~ 1,
    age==1 ~ 4,
    TRUE ~ 5
  ),
  ax = case_when(
    age==0 ~ 0.07 + 1.7*Mx,
    age==1 ~ 1.5,
    age==90 ~ 1/Mx,
    TRUE ~ 2.5
  ),
  qx = n * Mx / (1 + (n - ax)* Mx), # probability of death
  px = 1 - qx, # probability of survival
```

```

lx = lag(cumprod(px), default = 1), # l(x+n) = lx*px
dx = lx - lead(lx, default = 0), # ndx = lx - l(x+n)
Lx = n * lead(lx, default = 0) + (ax* dx), # nLx = n*l(x+n) + nax*ndx
Tx = rev(cumsum(rev(Lx))), # sum of Lx above age x
ex = Tx / lx
)

```

```

dlt_alternative <- dl |>
  mutate(n = case_when(
    age==0 ~ 1,
    age==1 ~ 4,
    TRUE ~ 5
  ),
  ax = case_when(
    age==0 ~ 0.07 + 1.7*Mx,
    age==1 ~ 1.5,
    age==90 ~ 1/Mx,
    TRUE ~ 2.5
  ),
  qx = n * Mx / (1 + (n - ax)* Mx), # probability of death
  px = 1 - qx)

dlt_alternative$lx <- NA
dlt_alternative$lx[1] <- 1

for(i in 2:nrow(dlt_alternative)){
  dlt_alternative$lx[i] <- dlt_alternative$lx[i-1]*dlt_alternative$px[i-1]
}

dlt_alternative$dx <- NA
for(i in 1:nrow(dlt_alternative)){
  dlt_alternative$dx[i] <- dlt_alternative$lx[i] - dlt_alternative$lx[i+1]
}

dlt_alternative$Lx <- NA

for(i in 1:nrow(dlt_alternative)){
  dlt_alternative$Lx[i] <- dlt_alternative$n[i]*dlt_alternative$lx[i+1] +dlt_alternative$ax[i]
}

```

What's the life expectancy at age 10?

Exercise

Calculate the lifespan disparity for Canada in 2016.

```
d_lt |>
  mutate(prod = dx*ex) |>
  summarize(lifespan_disparity = sum(prod))
```

```
# A tibble: 1 x 1
  lifespan_disparity
          <dbl>
1             12.5
```

```
dl2 <- d |>
  filter(year==2012|year==2016, cause=="Total, all causes of death [A00-Y89]") %>%
  mutate(age = start_age, Mx = rate/100000) %>%
  select(year, age, Mx)
```

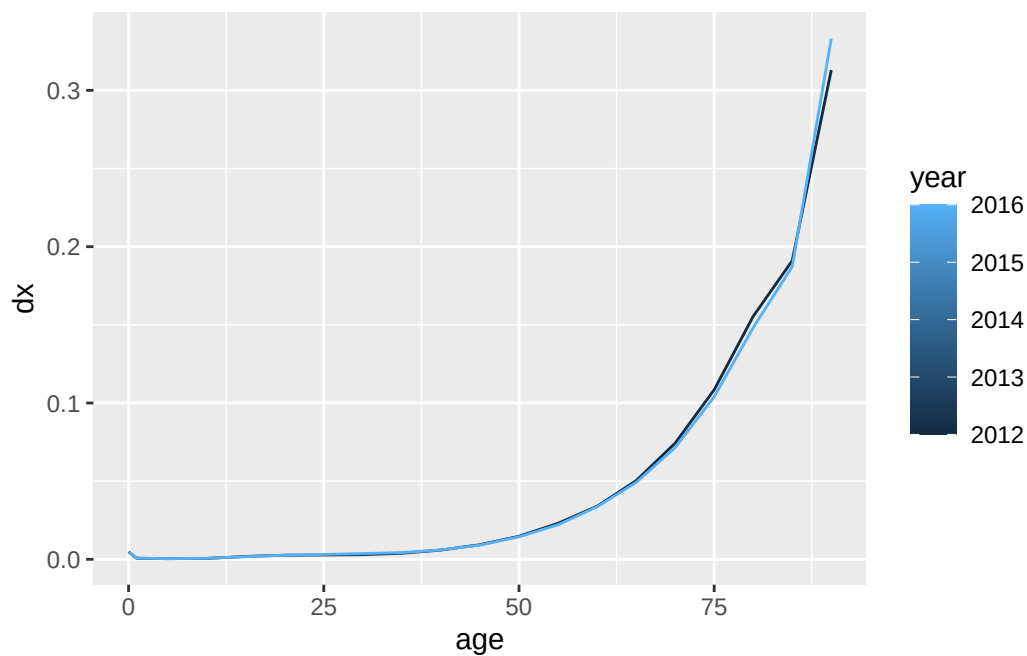
```
d_lt2 <- dl2 |>
  group_by(year) |>
  mutate(n = case_when(
    age==0 ~ 1,
    age==1 ~ 4,
    TRUE ~ 5
  ),
  ax = case_when(
    age==0 ~ 0.07 + 1.7*Mx,
    age==1 ~ 1.5,
    age==90 ~ 1/Mx,
    TRUE ~ 2.5
  ),
  qx = n * Mx / (1 + (n - ax)* Mx), # probability of death
  px = 1 - qx, # probability of survival
  lx = lag(cumprod(px), default = 1), # l(x+n) = lx*px
  dx = lx - lead(lx, default = 0), # ndx = lx - l(x+n)
  Lx = n * lead(lx, default = 0) + (ax* dx), # nLx = n*l(x+n) + nax*ndx
  Tx = rev(cumsum(rev(Lx))), # sum of Lx above age x
  ex = Tx / lx
  ) |>
  arrange(year)
```

```
# let's calculate lifespan disparity for years 2012 and 2016
```

```
d_lt2 |>  
  group_by(year) |>  
  mutate(prod = dx*ex) |>  
  summarize(ld = sum(prod))
```

```
# A tibble: 2 x 2  
  year    ld  
  <dbl> <dbl>  
1  2012  12.2  
2  2016  12.5
```

```
d_lt2 |>  
  ggplot(aes(age, dx, color = year, group = year))+  
  geom_line()
```



Calculate cause-deleted life expectancy

Now calculate life expectancy if all intentional injuries were deleted. Get the data we need:

```
dls <- d %>%
  filter(year==2016, cause=="Total, all causes of death [A00-Y89]"|cause=="Intentional self-harm (suicide) [X60-X84, Y87.0]")
  mutate(age = start_age, Mx = rate/100000) %>%
  select(age, cause, Mx) %>%
  mutate(cause = ifelse(cause=="Intentional self-harm (suicide) [X60-X84, Y87.0]", "suicide", "Total, all causes of death [A00-Y89]"))
  spread(cause, Mx) %>%
  rename(Mx_i = suicide,
         Mx = total)

head(dls)
```

```
# A tibble: 6 x 3
   age    Mx_i    Mx
<dbl> <dbl> <dbl>
1     0 0      0.0045
2     1 0      0.00019
3     5 0      0.00008
4    10 0.00003 0.00013
5    15 0.00009 0.00033
6    20 0.00012 0.00053
```

Exercise: Cause-deleted life expectancy

You need to create the same columns as above, but with the cause-deleted versions ($-i$). Do this by first creating the ratio $R_x^{-i} = \frac{M_x^{-i}}{M_x}$, use this to get ${}_nq_x^{-i}$, and the rest is the same.

What's the cause-deleted life expectancy at age 10? What's the implied life lost due to suicide?

```
dls <- dls |>
  mutate(Rx_minus_i = (Mx-Mx_i)/Mx)
```

```
lt_cd <- dls |>
  mutate(n = case_when(
    age==0 ~ 1,
    age==1 ~ 4,
    TRUE ~ 5
  ),
  ax = case_when(
    age==0 ~ 0.07 + 1.7*Mx,
    age==1 ~ 1.5,
```

```

    age==90 ~ 1/Mx,
    TRUE ~ 2.5
  ),
  qx = n * Mx / (1 + (n - ax)* Mx), # probability of death
  qx_minusi = qx*Rx_minusi,
  px = 1 - qx, # probability of survival
  lx = lag(cumprod(px), default = 1), # l(x+n) = lx*px
  dx = lx - lead(lx, default = 0), # ndx = lx - l(x+n)
  Lx = n * lead(lx, default = 0) + (ax* dx), # nLx = n*l(x+n) + nax*ndx
  Tx = rev(cumsum(rev(Lx))), # sum of Lx above age x
  ex = Tx / lx,
  ## CAUSE DELETED COLUMNS
  px_minusi = 1 - qx_minusi, # probability of survival
  lx_minusi = lag(cumprod(px_minusi), default = 1), # l(x+n) = lx*px
  dx_minusi = lx_minusi - lead(lx_minusi, default = 0), # ndx = lx - l(x+n)
  Lx_minusi = n * lead(lx_minusi, default = 0) + (ax* dx_minusi), # nLx = n*l(x+n) + nax*
  Tx_minusi = rev(cumsum(rev(Lx_minusi))), # sum of Lx above age x
  ex_minusi = Tx_minusi / lx_minusi
)

lt_cd |>
  select(age, ex, ex_minusi)

```

```

# A tibble: 20 x 3
   age    ex ex_minusi
  <dbl> <dbl>   <dbl>
1     0 82.3     82.6
2     1 81.6     81.9
3     5 77.7     78.0
4    10 72.7     73.0
5    15 67.8     68.1
6    20 62.9     63.1
7    25 58.0     58.3
8    30 53.2     53.4
9    35 48.4     48.6
10   40 43.6     43.7
11   45 38.8     39.0
12   50 34.2     34.3
13   55 29.7     29.7
14   60 25.3     25.4
15   65 21.2     21.2
16   70 17.3     17.3

```

17	75	13.6	13.6
18	80	10.4	10.4
19	85	7.60	7.60
20	90	5.46	5.46

END

DO NOT DO!!!! MONICA GOT TOO EXCITED!!!

Decomposition

Let's read in WPP data from last week and calculate the age-specific mortality rates:

```
d_male <- read_xlsx(here("data/WPP2024_POP_F01_2_POPULATION_SINGLE_AGE_MALE.xlsx"), skip = 16)
d_male$sex <- "Male"
d_male <- d_male |> drop_na(Year)
d_female <- read_xlsx(here("data/WPP2024_POP_F01_3_POPULATION_SINGLE_AGE_FEMALE.xlsx"), skip = 16)
d_female$sex <- "Female"
d_female <- d_female |> drop_na(Year)

d <- rbind(d_male, d_female)
rm(d_male, d_female)

d <- d |>
  clean_names() |>
  select(region_subregion_country_or_area, iso3_alpha_code, year, x0:sex) |>
  rename(region = region_subregion_country_or_area) |>
  mutate(across(x0:x100, as.numeric))

d_male <- read_xlsx(here("data/WPP2024_MORT_F01_2_DEATHS_SINGLE_AGE_MALE.xlsx"), skip = 16)
d_male$sex <- "Male"
d_male <- d_male |> drop_na(Year)
d_female <- read_xlsx(here("data/WPP2024_MORT_F01_3_DEATHS_SINGLE_AGE_FEMALE.xlsx"), skip = 16)
d_female$sex <- "Female"
d_female <- d_female |> drop_na(Year)

dm <- rbind(d_male, d_female)
rm(d_male, d_female)

dm <- dm |>
  clean_names() |>
```



```

select(region_subregion_country_or_area, iso3_alpha_code, year, x0:sex) |>
rename(region = region_subregion_country_or_area) |>
mutate(across(x0:x100, as.numeric))

d_long <- d |>
pivot_longer(x0:x100, names_to = "age", values_to = "pop") |>
mutate(age = as.numeric(str_remove(age, "x")))

dm_long <- dm |>
pivot_longer(x0:x100, names_to = "age", values_to = "deaths") |>
mutate(age = as.numeric(str_remove(age, "x")))

# join these two tibbles and calculate rates

asmr <- d_long |>
left_join(dm_long) |>
mutate(mx = deaths/pop)

```

Do the decomposition of the difference between Kenya and Canada:

```

asmr |>
filter(region == "Kenya", year == 2023) |>
select(sex, age, pop, mx) |>
rename(pop_kenya = pop, mx_kenya = mx) |>
left_join(asmr |>
  filter(region == "Canada", year == 2023) |>
  select(sex, age, pop, mx) |>
  rename(pop_can = pop, mx_can = mx) ) |>
mutate(prop_kenya = pop_kenya/sum(pop_kenya),
  prop_can = pop_can/sum(pop_can)) |>
mutate(rate_diff = mx_kenya - mx_can,
  prop_diff = prop_kenya - prop_can) |>
mutate(ave_rate = (mx_kenya+mx_can)/2,
  ave_prop = (prop_kenya+prop_can)/2) |>
mutate(age_contr = prop_diff*ave_rate,
  rate_contr = rate_diff*ave_prop) |>
summarize(age_total_contr = sum(age_contr),
  rate_total_contr = sum(rate_contr)) |>
mutate(total_diff = age_total_contr+rate_total_contr)

```

A tibble: 1 x 3

	age_total_contr	rate_total_contr	total_diff
	<dbl>	<dbl>	<dbl>
1	-0.0107	0.00998	-0.000724

Check that the difference is actually the difference between the two CDRs

```
asmr |>
  filter(region == "Kenya"|region=="Canada",year==2023) |>
  group_by(region) |>
  summarize(cdr = sum(mx*pop)/sum(pop)) |>
  summarise(diff = cdr[region=="Kenya"] - cdr[region=="Canada"])
```

```
# A tibble: 1 x 1
  diff
  <dbl>
1 -0.000724
```

Exercise

Decompose the difference in CDRs between USA and Japan in the year 2023. Is the majority of the difference due to age structure or mortality?